

Peoplesoft ID:



EEE3094S 2023: Exam

- The exam is open book
- Answer on the exam paper. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write "see attached pages" by the question) and make sure they are marked with your peoplesoft ID. There is also an extra page of writing space provided at the end of the paper.
- Additional copies of all plots are included at the end of the paper in case you make a mistake on the first one. Please note near the question that you have used the extra figure. You may write on your figures in pencil, but please ensure that the pencil is sharp and dark enough to be easily legible and does not smudge.
- There are five questions totalling 100 marks.
- You have 15 minutes reading time and 3 hours to complete the exam.
- The last page is an information sheet.

Question 1 [15]

The Bode plot on the following page shows the frequency response of a system that is stable in open loop. Based on this response, decide whether each of the following statements about this system are TRUE or FALSE.

You may wish to draw on or annotate the plot to assist your working or draw attention to key features.
[5 x 3 marks]

- a) The open-loop response of the system is oscillatory.
- b) The system is greater than second-order.
- c) The system has a gain margin of -12 dB.
- d) The system will track a position input without error.
- e) The open-loop system will attenuate (decrease) noise.

a)

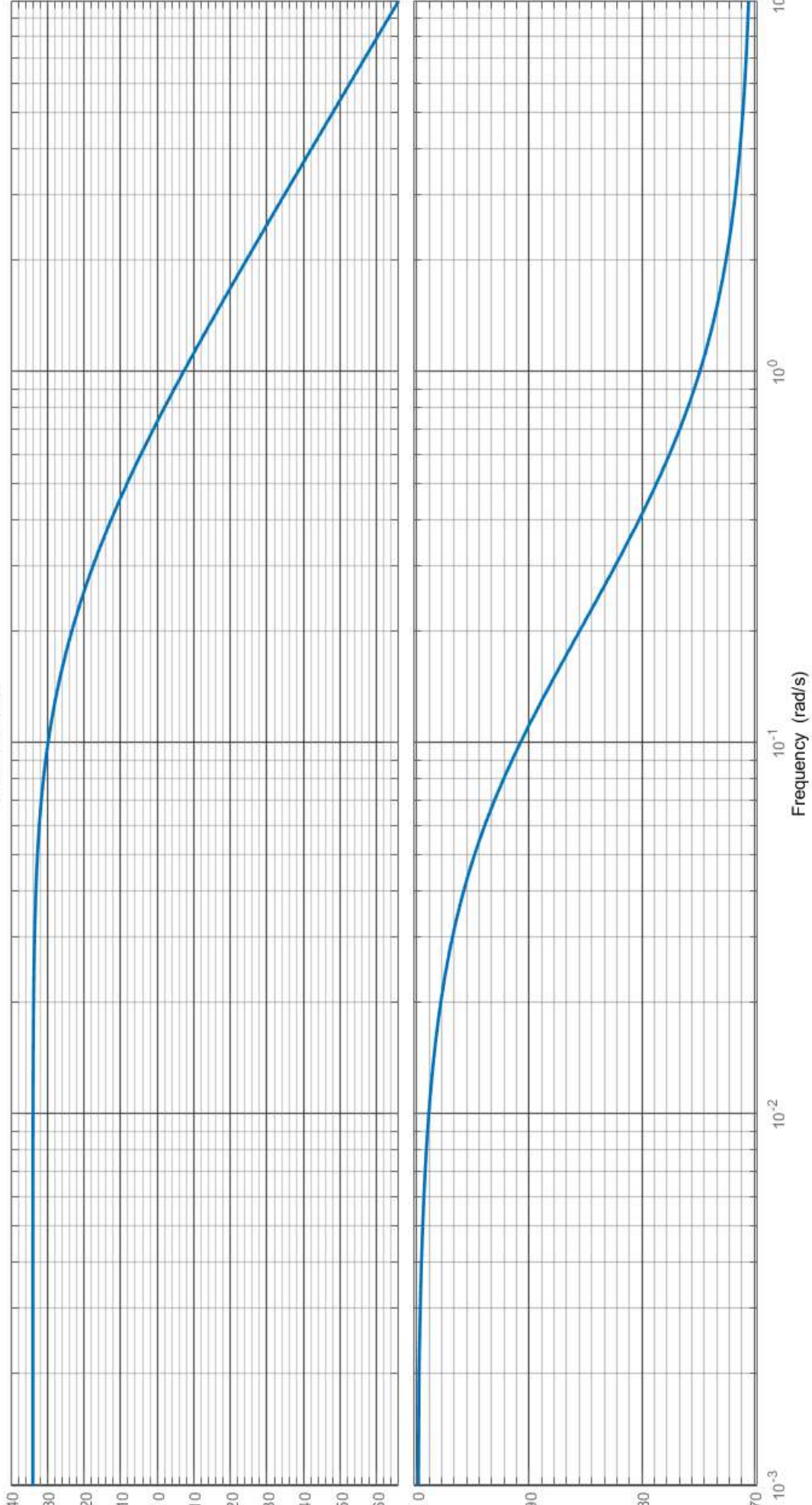
b)

c)

d)

e)

Bode Diagram



Question 2 [15]

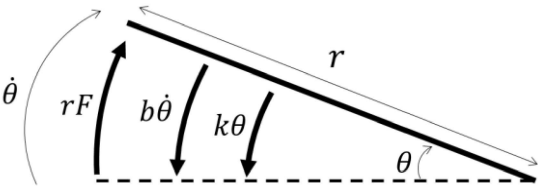
Wouldn't it be cute if, instead of a regular cat flap, you had a pair of little doors that swing out sideways so that your cat can burst through them like an outlaw barging into the one saloon in a little Wild West town to cause a ruckus? Let's design some.

Let the angle of the door with respect to its closed position be θ . The door must open bi-directionally, so θ can be positive or negative.

We need a torsional spring to pull the doors back to the closed position automatically once they have been opened. Let the spring constant be k .

We also need something to damp the motion of the doors, so they don't swing wildly and noisily every time your cat passes through them during their nighttime excursions.

The cat will exert a force F on the door. The cat passing through will deliver a short push, like an impulse, so what we are trying to do is design the open-loop impulse response of the doors to give us the fastest possible closing time without repeated swings (overshoot).



You 3D print little saloon doors that are 7 cm wide by 15 cm high, and weigh 50 g. This gives a moment of inertia of $437.5 \frac{g}{cm^2}$. You then find damped hinges with a damping coefficient of 150 (g/s). The only remaining design parameter is the spring constant.

- a) Write a differential equation modelling the motion of the cat door in response to the force exerted by the cat, and hence, derive a transfer function model. [3]
- b) Draw a root locus representing the open-loop pole locations for the changing spring constant. [4]
- c) What damping condition meets the requirements of "the fastest possible closing time without repeated swings"? Determine the location of the closed-loop poles when this condition is met, and hence, the spring constant required to achieve it. [3]
- d) The torsional springs available at your nearest hardware store have the coefficients (in g/s^2)
(A) 0.1
(B) 10
(C) 50
(D) 100

Select the one that best matches your calculations. Determine what the closed-loop poles will be, and how the response will differ from the ideal condition of fastest settling time with no overshoot (quantify your response in terms of a difference in settling time to 2% or percentage overshoot). [5]

a)

b)

b)

c)

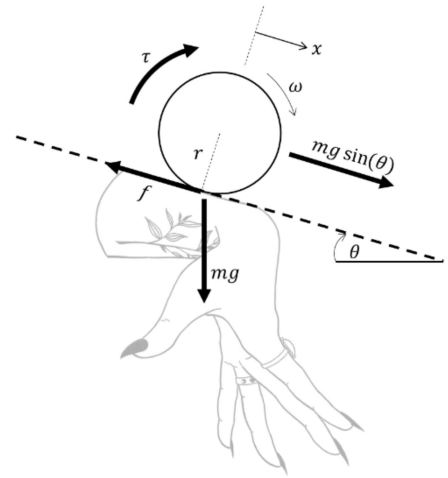
d)

Question 3 [30]

Contact juggling is a form of juggling where a ball (or another object, such as a staff) is rolled along the body rather than being thrown into the air. It is a fun hobby practiced exclusively by interesting and attractive people.

An important contact juggling skill is the *stall*. This involves stopping and balancing the ball on some part of the body.

The diagram to the right shows the forces acting on a sphere as it rolls on an arm inclined at an angle θ . During the stall, we can assume that the angle will be small, and therefore, that $\sin(\theta) \approx \theta$.



- a) Show that the transfer function $G(s) = \frac{5g/7}{s^2}$ approximates the relationship between the input angle $\theta(t)$, and output $x(s)$ – the distance from the stall point parallel to the arm. [6]

You may consider the following assumptions in your analysis:

- The sphere is uniform and solid, having a mass moment of inertia $\frac{2}{5}mr^2$, where m is the mass and r is the radius.
- The friction force f is sufficient that the sphere rolls without slipping, therefore the linear acceleration $\ddot{x}$ and angular acceleration $\dot{\omega}$ are related by $\dot{\omega} = \frac{\ddot{x}}{r}$.
- The sphere is initially at rest at the stall point.

Hints:

- Model the x and rotational directions separately, and then combine them into a single differential equation.
- The torque τ on the sphere is produced by the friction force.

- b) Based on this transfer function, would you consider the stall to be a stable point of equilibrium? [2]
- c) The juggler's senses (proprioception, vision) form a negative feedback loop that allows them to keep the ball at the stall point. Would it be possible for the juggler to keep the ball steady if they are only aware of its position (i.e. without considering the velocity)? Explain. You may wish to support your explanation with a sketch of the root locus. [5]

By analyzing the juggler's response to errors in the sphere's position, it is determined that the angle of the arm is a function of the position error $e(s)$, rate of change of the error $\dot{e}(s)$ (i.e. the velocity error) and the rate of change of the arm angle $\dot{\theta}(s)$:

$$\theta(t) = Ke(t) + 5K\dot{e}(t) + 20\dot{\theta}(t)$$

where K is a gain. This corresponds to a controller $C(s) = \frac{K(s+5)}{s+30}$.

- d) Sketch a root locus for the controlled system. [5]
- e) Is it possible to bring the sphere to a standstill without oscillation? [2]
- f) Is it possible to bring the sphere to a standstill in less than 0.3 seconds? You may use the 5% settling time as your reference. [2]
- g) Indicate the region on your root locus where the dominant poles must lie to produce a closed-loop settling time (5%) of less than 0.5 seconds, and less than 20% overshoot. [4]

- h) Use the angle criterion to show that the poles $-6.1 \pm j6.98$ are on the root locus of the controlled system. Determine the gain K required for these to be the dominant closed-loop poles. [4]

a)

b)

c)

c)

d)

d,g)

e)

f)

g)

[illegible][illegible]

Question 4 [30]

The Bode plot and inverse Nichols chart on the next pages show the frequency response of a voltage-controlled sun lamp. You wish to place this lamp in feedback to maintain a constant light level to grow plants indoors.

This goal is complicated by output disturbances in the form of ambient light shifts. You are also using a second-hand lamp where the brightness at a given voltage may vary from the values stated by the manufacturer due to wear on the filament. Your analysis should therefore allow for a fluctuation of 1 dB in the gain of the system.

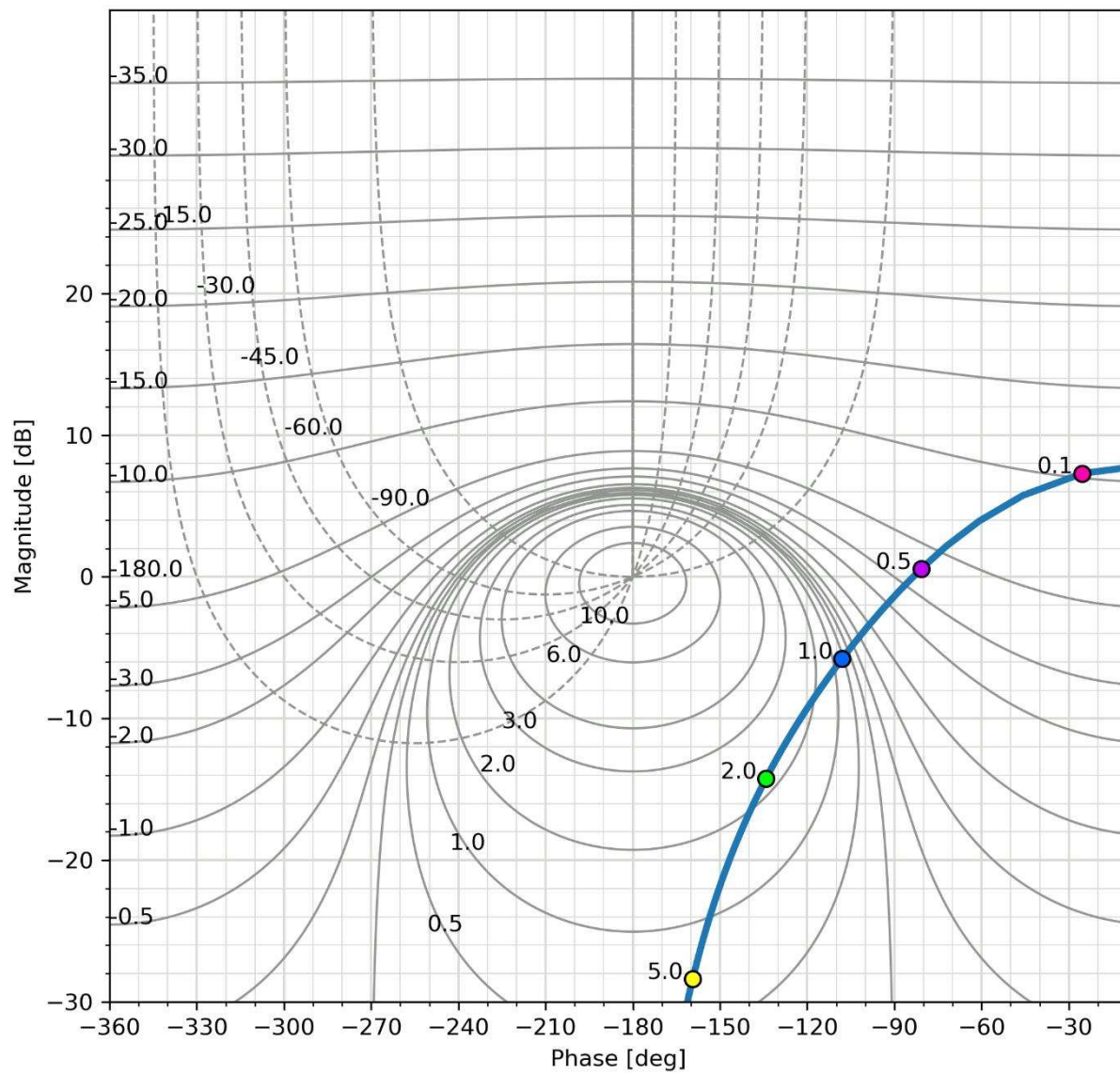
You may assume that your light sensor is perfectly accurate.

Your priorities for the design are that it be:

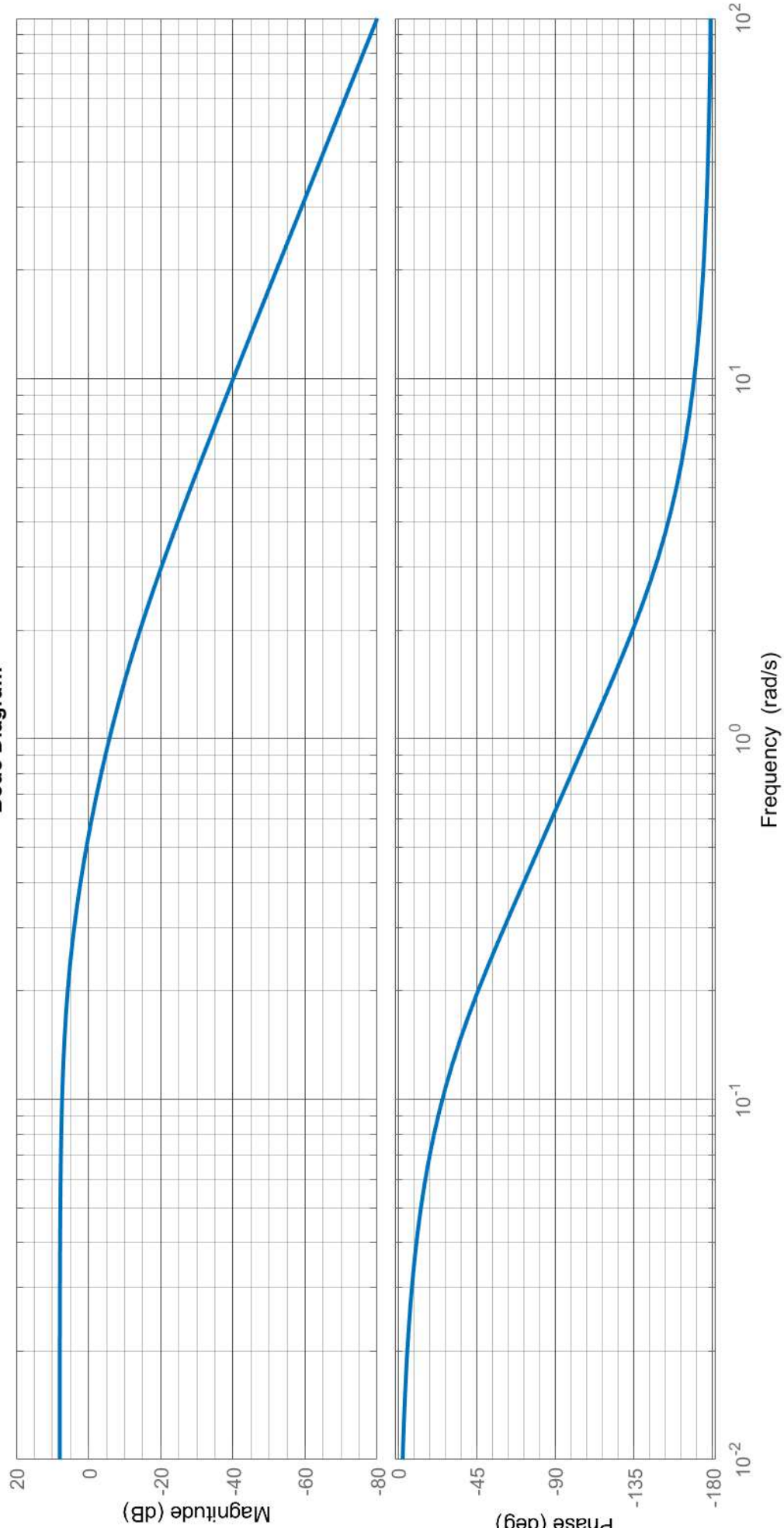
- 1) Robust (sensitivity ≤ 3 dB at all frequencies)
- 2) Accurate: it must maintain the ideal light level with less than 5% error.
- 3) Well-damped (the light level must not overshoot the ideal by more than 10%, as this would risk scorching the plants)
- 4) Simple
- 5) Inexpensive

Note: You may annotate your charts in pencil, but please ensure that it is dark enough to be clearly legible. Spare charts are attached on the following page if required (cross out the first chart to indicate that you are using the spare).

- a) What effect will the ambient light changes have on the uncontrolled closed-loop system? You can model this as a constant disturbance. Give your answer as a percentage of the magnitude of the disturbance. [4]
- b) Calculate the minimum low-frequency gain required of the system, and hence, how much the gain must be increased by. Indicate this specification on the inverse Nichols chart. [6]
- c) Comment on the advantages and disadvantages of implementing this gain increase using a proportional controller. Would you recommend this type of control in this case? Support your answer by sketching the response of the controlled system on the inverse Nichols chart. [4]
- d) You decide to implement a **lag compensator** to control the system. Justify this decision by explaining the advantages and disadvantages of this type of controller compared to (i) an integral controller, and (ii) a lead compensator. Ensure that your explanation considers the specific priorities of this design. [6]
- e) Indicate the damping and robustness specifications on the inverse Nichols chart. [4]
- f) Design the lag compensator. [6]



Bode Diagram



a)

b)

c)

d.i)

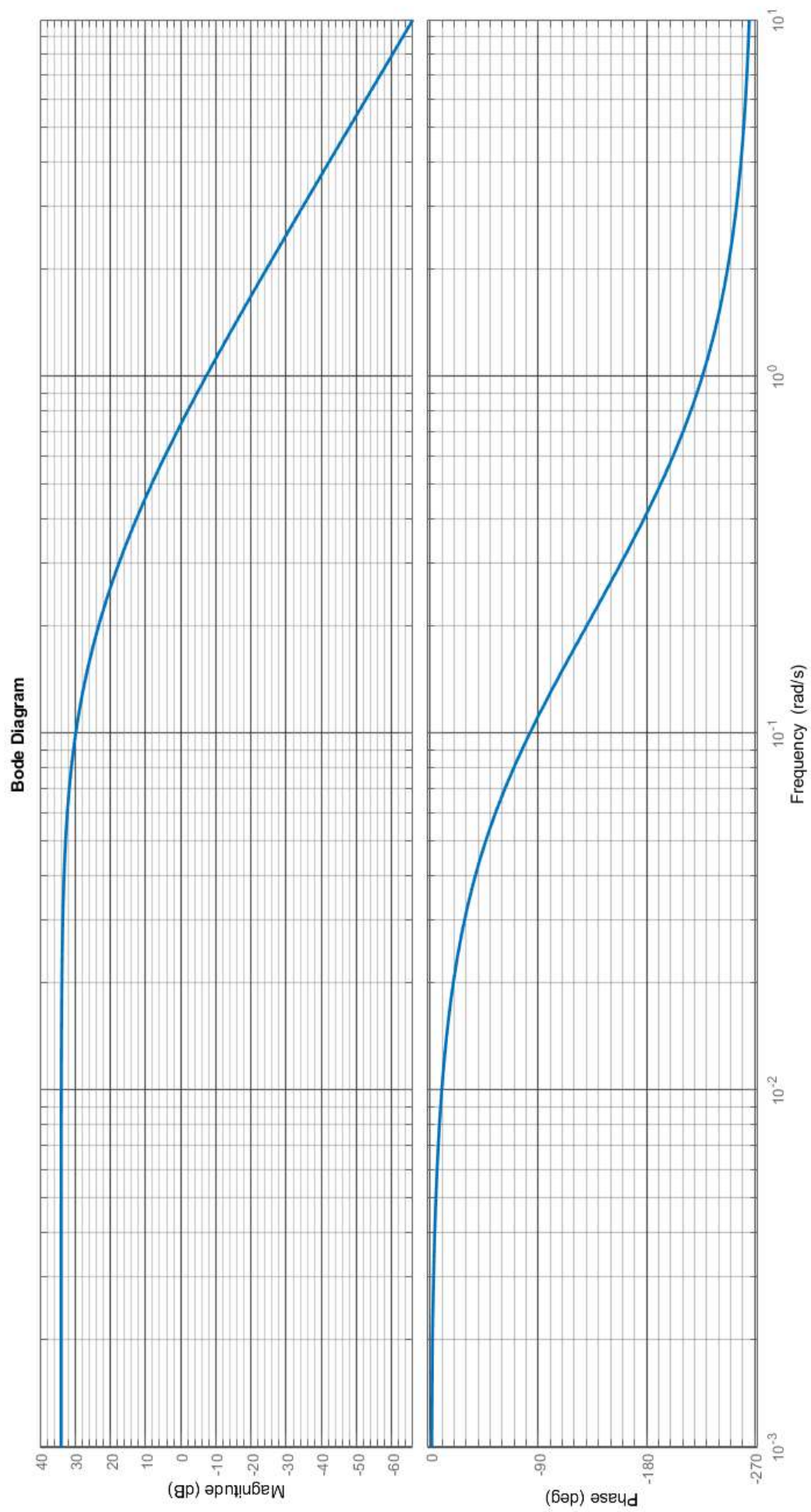
d.ii)

e)

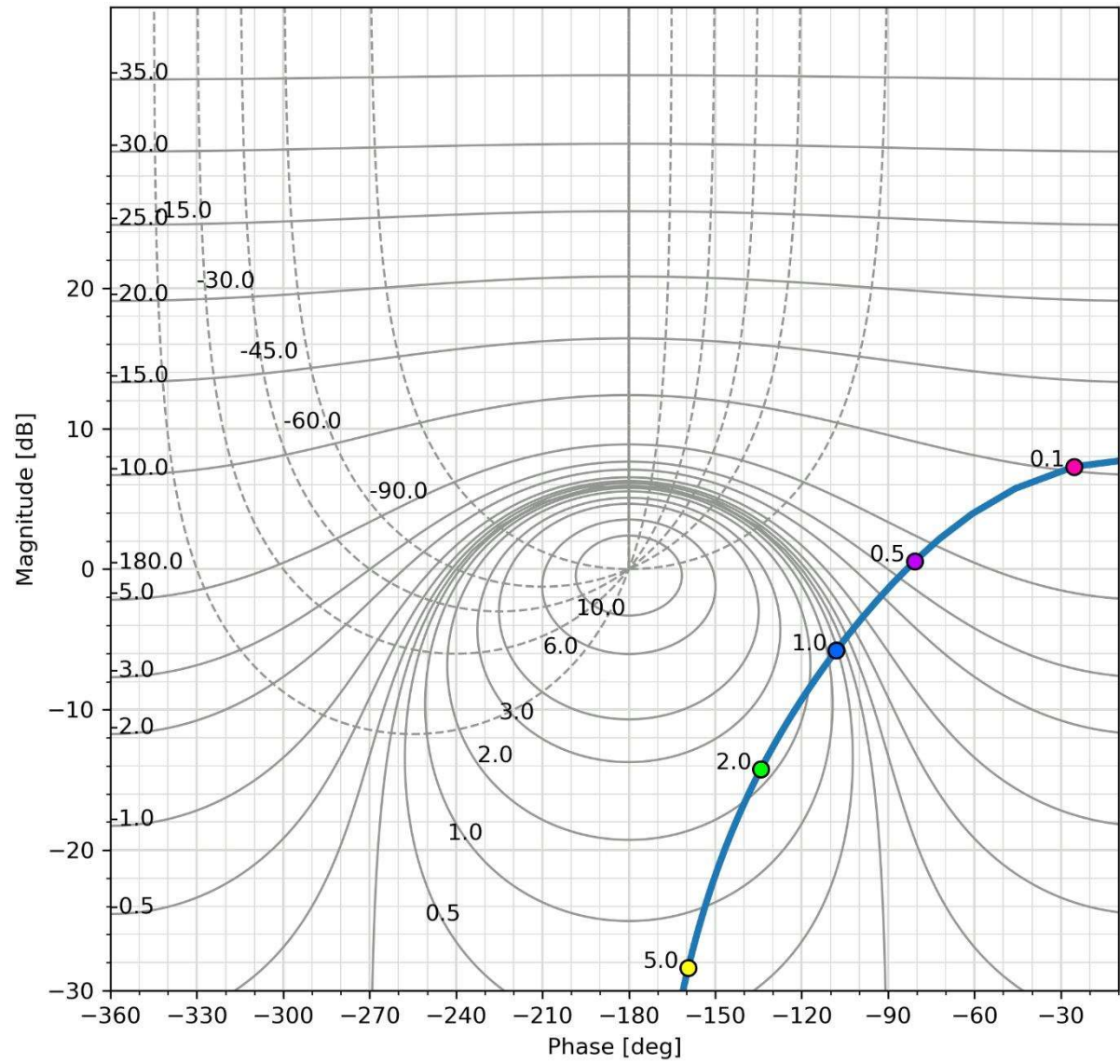
f)

[illegible]

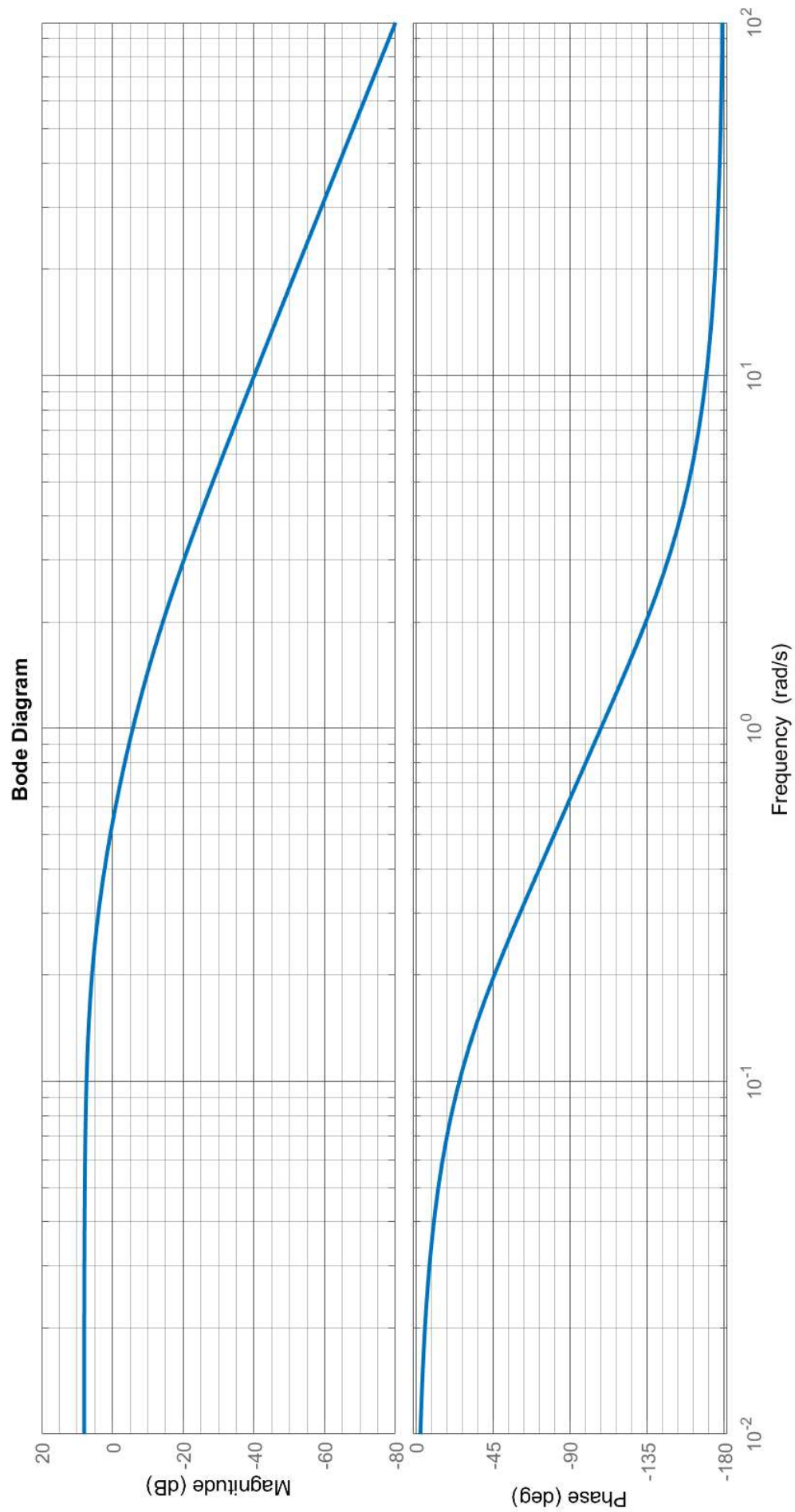
Question 1 – Extra Bode Plot



Question 4 - Extra Inverse Nichols Chart



Question 4 – Extra Bode Plot



Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} \right)}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	cosinusoid
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>